Reg. No. | | | | | | | | | | [12EC835]

# EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE - 2016

## OPERATING SYSTEMS & RTOS

**Time: 3 Hours** **Maximum Marks:70**

**Instructions:**

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

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|---|---|---|---|
| 1. | a. | Explain the architecture of the following operating systems with neat block diagrams.<br>i. Monolithic Operating System<br>ii. Microkernel Operating System | 12 |
| | b. | What are the objectives of operating system? | 02 |
| 2. | a. | Write the differences between long-term medium term and short term schedulers. | 10 |
| | b. | With neat relevant diagrams, explain briefly hard and soft real time systems. | 04 |
| 3. | a. | Explain race condition in interprocess communication with a spooler directory example. And also write necessary and sufficient conditions to implement critical section/region in IPC. | 12 |
| | b. | List the differences between process and threads. | 02 |
| 4. | | What do you mean by deadlock? Write a short note on deadlock prevention. | 14 |
| 5. | a. | Explain paging technique in memory management. | 08 |
| | b. | Explain briefly different types of memory allocation mechanism used by operating systems. | 04 |
| | c. | List the differences between logical and physical address space. | 02 |
| 6. | a. | With the neat relevant diagrams, explain page-in and page-out operations. | 08 |
| | b. | What are the advantages and disadvantages of memory segmentation? | 03 |
| | c. | Compare memory segmentation and paging mechanism. | 03 |
| 7. | a. | What do you mean by I/O buffering and also describe the following terminologies with neat relevant diagrams.<br>i. Un buffered input.<br>ii. Buffering in user space<br>iii. Double buffering in the kernel | 10 |
| | b. | Explain briefly different mechanisms to access files. | 04 |

8. a. Explain various components of Linux Operating system with architecture diagram and also mention the features and draw backs of Linux OS. **08**

b. Write a shell script to get the following output pattern. **04**

$./abc 6 7

The sum of 6 + 7 = 13

The difference of 7 and 6 =1

The product of 7 and 6 = 42

The factorial of 6 = 720

The script file name is abc.

c. Explain briefly the use of "chmod" command in Linux OS. **02**